# Comparison of Scintillation Crystals

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**Abstract**

This experiment aimed to measure and compare the decay times and light yields of three scintillating crystals: BGO, CsI(Tl), and LYSO(Ce). Using a Cs-137 source to excite the crystals, the emitted photons were detected by a Silicon Photomultiplier (SiPM), and the signals were digitized using an ADC unit. We calculated the average waveform and light yield for each crystal. The measured decay times were 67.75 ns for LYSO(Ce), 307.32 ns for BGO, and 935.14 ns for CsI(Tl). These results were compared with accepted values of 40 ns for LYSO(Ce), 300 ns for BGO, and 1000 ns for CsI(Tl), showing deviations of approximately 69.38%, 2.44%, and 6.48%, respectively [1]. Experimental limitations, such as equipment constraints and signal processing methods, likely contributed to the discrepancies observed. Improved alignment and analysis techniques led to more accurate measurements compared to initial results.

## 1 Introduction

The study of scintillating crystals, such as Bismuth Germanate (BGO), Thallium-doped Cesium Iodide (CsI(Tl)), and Cerium-doped Lutetium Yttrium Orthosilicate (LYSO(Ce)), is crucial for applications in medical physics, particle physics, and radiation detection [1]. These crystals are known for their ability to convert gamma rays into visible light through scintillation, a process that can be detected by a photodetector like a Silicon Photomultiplier (SiPM) [2]. In this experiment, we aim to measure the decay times and light yields of the three crystals when exposed to gamma radiation from a Cs-137 source.

Cesium-137 is a radioactive isotope that decays by beta emission, producing gamma rays with an energy of 662 keV. These gamma rays interact with matter through three main processes: the photoelectric effect, Compton scattering, and pair production. The dominant interaction depends on the energy of the gamma rays and the atomic number of the material. For lower-energy gamma rays like those from Cs-137, the photoelectric effect typically dominates.

In the photoelectric effect, a gamma photon is completely absorbed by an atom, leading to the ejection of a photoelectron [3]. The energy balance of this process is given by:

$$E_\gamma = E_{\text{binding}} + K_e, \tag{1}$$

where $E_\gamma$ is the energy of the incoming gamma photon, $E_{\text{binding}}$ is the binding energy of the electron in the atom, and $K_e$ is the kinetic energy of the emitted photoelectron.

Compton scattering occurs when a gamma photon scatters off a free or loosely bound electron, resulting in a photon of reduced energy and a recoiling electron [4]. The change in wavelength of the photon is described by the Compton equation:

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta), \tag{2}$$

or in terms of energy:

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2}(1 - \cos\theta)}, \tag{3}$$

where $\lambda$ and $\lambda'$ are the initial and final wavelengths of the photon, $h$ is Planck's constant, $m_e$ is the electron mass, $c$ is the speed of light, $\theta$ is the scattering angle, $E_\gamma$ is the initial photon energy, and $E'_\gamma$ is the scattered photon energy.

Pair production, which requires gamma rays with energy greater than 1.022 MeV (twice the rest mass energy of an electron), results in the creation of an electron-positron pair. Since Cs-137 has an average gamma ray energy of 0.662 MeV, pair production is not relevant in this experiment [5].

Scintillating crystals are essential in detecting these interactions because they convert the gamma rays into visible light, which can then be measured and analyzed to determine important properties such as decay time and light yield. These properties provide insights into the efficiency and effectiveness of the crystals in radiation detection. BGO, CsI(Tl), and LYSO(Ce) crystals, in particular, are widely used in medical imaging devices such as PET scanners and in security systems for radiation detection.

This experiment is significant in exploring how these crystals react when exposed to gamma rays, which is critical for developing technologies that rely on scintillation for detecting and analyzing radiation. By measuring the decay time and light yield of each crystal, we aim to evaluate their suitability for various applications and to better understand their physical properties.

# 2 Experimental Procedure

## 2.1 Equipment

The following equipment was used for the experiment are as follows and labeled in Fig 1:

- **SP500 Power Supply and Amplification Unit (PSAU):** Provides the necessary voltage (set to 55 V) and amplifies the signal received from the SiPM.
- **SP506 Mini-Spectrometer with SiPM:** Contains the SiPM that detects photons emitted by the scintillating crystals.
- **Desktop Digitizer (ADC unit):** Converts the analog signal from the SiPM into digital data for further analysis.
- **Cs-137 Source:** Used to excite the scintillating crystals by emitting gamma rays at 662 keV.
- **Computer with Windows OS:** Connected to both the PSAU and digitizer, allowing us to control the experiment using CAEN software.

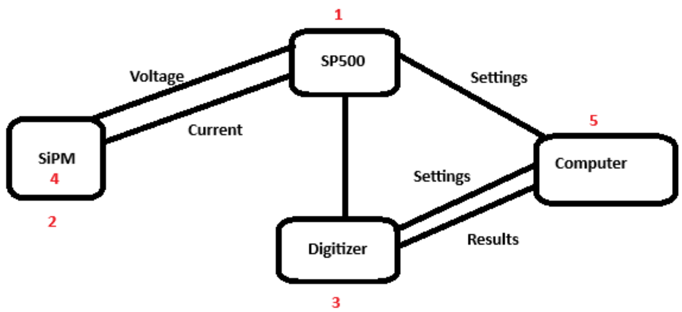


Figure 1: (1) : SP500 Power Supply and Amplification Unit (PSAU); (2) : SP506 Mini-Spectrometer with SiPM; (3) : Desktop Digitizer (ADC unit); (4) : Cs-137 Source; (5): Computer with Windows OS

The software used included **HERA**, which controls the PSAU settings including the gain, and **COMPASS**, which controls the ADC unit settings, particularly the discriminant level.

## 2.2 Experimental Setup

In the experimental setup, the scintillating crystal was placed face down on the SiPM. The Cs-137 source was positioned on top of the crystal; gamma rays emitted from the source excited the crystal, producing scintillations. Photons produced by the crystal were detected by the SiPM, which converted the photon signals into an electric current and amplified them with the SP500 PSAU. The amplified signal was then digitized by the ADC unit and sent to the connected computer for data acquisition. The data was collected as ADC channels in 4 ns intervals.

## 2.3 Equipment Settings

Proper settings of the equipment were crucial for accurate data collection.

**Gain Settings:**

The gain in the PSAU was adjusted for each crystal to prevent signal saturation and optimize the signal-to-noise ratio. The gain values for each crystal were as follows:

Table 1: Gain and Threshold Settings for Each Crystal

| Crystal | Gain (dB) | Threshold (lsb) | Measurement Time (s) |
|---|---|---|---|
| CsI(Tl) | 20 | 50 | 45 |
| BGO | 32 | 50 | 45 |
| LYSO(Ce) | 20 | 50 | 45 |

The gain formula used in the data analysis was:

$$\text{Adjusted Signal} = \text{Signal} \times 10^{-\frac{\text{Gain (dB)}}{20}} \tag{4}$$

**Threshold (Discriminant Level):**

The threshold was set to filter out noise and low-energy signals that could overwhelm the software. Based on the dark count measurements, a threshold of 50 lsb (least significant bits) was used for all crystals.

**Voltage Settings:**

The SiPM was supplied with a voltage of 55 V. Slight fluctuations of ±0.5 V were observed, which could introduce small errors.

## 2.4 Data Acquisition

Data was collected using the COMPASS software. Each run lasted approximately 45 seconds, capturing around 70,000 signals per run. The signals were recorded as voltage values for each ADC channel in 4 ns intervals and stored in CSV format.

## 2.5 Calibration Procedures

A dark count measurement was taken without any crystal or Cs-137 source to establish a baseline for noise. In order to account for the noise, we took $10 \cdot \sigma$ to be our threshold to ensure no noise. Our chosen value was 26 lsb with a gain of 32db, The standard deviation of the dark count was used to set the threshold level to be 10 times the standard deviation to ensure that random noise did not affect the measurements.

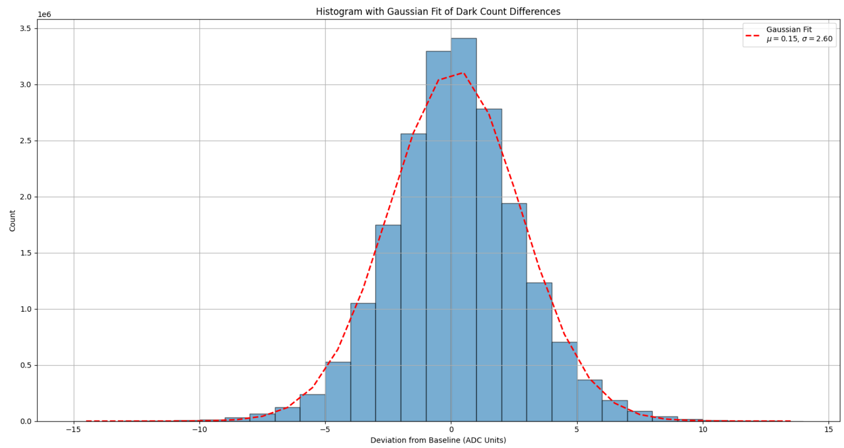


Figure 2: Dark count measurement showing the distribution of baseline noise.

### 2.6 Issues Encountered

Several issues were encountered during the experiment:

**Software Compatibility:**

The digitizer was initially incompatible with the HERA software. After several attempts over 10 hours, the issue was resolved by disabling the digitizer in HERA and using COMPASS software to control the ADC unit.

**Signal Saturation:**

Initial signal levels were too high, leading to saturation of the ADC unit. This was identified by observing flat-topped waveforms indicating clipping. The gain was lowered to prevent saturation.

**Low-Energy Signals Overwhelming Software:**

An excessive number of low-energy signals caused the software to crash. Increasing the threshold to 50 lsb helped filter out these signals.

**Equipment Limitations:**

Limitations in the PSAU prevented direct calculation of absolute light yields.

**Environmental Factors:**

Temperature fluctuations due to the experimental setup being near a window could have affected the measurements. To mitigate this, data on temperature conditions was collected in a single session to maintain consistent environmental conditions.

### 2.7 Data Collection and Management

Data files were named based on the crystal type and run number for organization. For example:

- `DataR_run_green_02.csv` for BGO (green)
- `DataR_run_white_02.csv` for CsI(Tl) (white)
- `DataR_run_black_02.csv` for LYSO(Ce) (black)

Data was processed using Python scripts, with code available on GitHub for reproducibility. Version control was managed using Git.

## 3 Data and Analysis

### 3.1 Data Collection

Data consisted of CSV files with approximately 70,000 rows per run and 250 columns representing voltage measurements at 4 ns intervals.

A sample of the data is shown below and in Fig 5:

Table 2: Sample Data Row

```
BOARD;CHANNEL;TIMETAG;ENERGY;SAMPLE_0;...;SAMPLE_250
0;0;2116000;117;21;0x0;1;3621;3621;3619;...;3620
```

Data was processed using Python scripts to:

- Adjust signals based on gain settings.
- Calculate baselines and subtract them from the signals.
- Sum voltage counts to determine light yields.
- Align and average waveforms for decay time analysis.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3610 | 3610 | 3611 | 3610 | 3611 | 3611 | 3611 | 3611 |
| 3611 | 3610 | 3610 | 3609 | 3611 | 3612 | 3610 | 3611 |
| 3611 | 3609 | 3610 | 3611 | 3609 | 3610 | 3611 | 3611 |
| 3614 | 3612 | 3613 | 3614 | 3612 | 3608 | 3611 | 3609 |
| 3611 | 3609 | 3609 | 3609 | 3606 | 3599 | 3592 | 3589 |
| 3612 | 3611 | 3612 | 3612 | 3612 | 3611 | 3612 | 3612 |
| 3607 | 3608 | 3606 | 3607 | 3608 | 3608 | 3609 | 3609 |
| 3605 | 3607 | 3606 | 3607 | 3608 | 3609 | 3607 | 3608 |
| 3609 | 3610 | 3609 | 3609 | 3609 | 3609 | 3610 | 3609 |
| 3612 | 3610 | 3611 | 3613 | 3612 | 3612 | 3610 | 3612 |
| 3612 | 3614 | 3613 | 3613 | 3613 | 3613 | 3613 | 3612 |
| 3609 | 3607 | 3610 | 3608 | 3609 | 3610 | 3609 | 3609 |
| 3604 | 3598 | 3587 | 3579 | 3570 | 3568 | 3565 | 3562 |
| 3545 | 3544 | 3542 | 3543 | 3541 | 3534 | 3536 | 3536 |
| 3607 | 3608 | 3608 | 3609 | 3611 | 3609 | 3609 | 3609 |
| 3611 | 3611 | 3610 | 3611 | 3610 | 3609 | 3611 | 3611 |
| 3610 | 3610 | 3610 | 3610 | 3610 | 3610 | 3609 | 3611 |
| 3605 | 3606 | 3608 | 3608 | 3606 | 3605 | 3602 | 3598 |

Figure 3: Data Sample from BGO CSV File

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3609 | 3607 | 3608 | 3610 | 3607 | 3610 | 3608 | 3608 |
| 3610 | 3609 | 3608 | 3611 | 3610 | 3610 | 3608 | 3609 |
| 3609 | 3609 | 3609 | 3611 | 3610 | 3609 | 3610 | 3610 |
| 3608 | 3607 | 3608 | 3608 | 3607 | 3608 | 3609 | 3608 |
| 3605 | 3606 | 3604 | 3603 | 3606 | 3606 | 3605 | 3606 |
| 3602 | 3604 | 3604 | 3604 | 3605 | 3604 | 3606 | 3605 |
| 3611 | 3611 | 3610 | 3612 | 3611 | 3612 | 3610 | 3612 |
| 3604 | 3606 | 3602 | 3602 | 3605 | 3605 | 3605 | 3605 |
| 3613 | 3614 | 3614 | 3614 | 3613 | 3613 | 3609 | 3605 |
| 3608 | 3608 | 3609 | 3609 | 3608 | 3607 | 3602 | 3593 |
| 3615 | 3613 | 3614 | 3614 | 3612 | 3616 | 3615 | 3614 |
| 3609 | 3608 | 3610 | 3610 | 3609 | 3610 | 3610 | 3610 |
| 3609 | 3607 | 3608 | 3609 | 3608 | 3609 | 3609 | 3609 |
| 3610 | 3608 | 3608 | 3611 | 3611 | 3610 | 3611 | 3610 |
| 3610 | 3610 | 3613 | 3611 | 3612 | 3612 | 3611 | 3611 |
| 3610 | 3611 | 3612 | 3612 | 3608 | 3610 | 3609 | 3608 |
| 3610 | 3608 | 3610 | 3609 | 3608 | 3610 | 3609 | 3612 |
| 3610 | 3609 | 3609 | 3609 | 3610 | 3610 | 3609 | 3610 |
| 3612 | 3610 | 3612 | 3612 | 3612 | 3613 | 3611 | 3612 |

Figure 4: Data Sample from LYSO CSV File

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 3609 | 3608 | 3609 | 3610 | 3609 | 3609 | 3608 | 3610 | 3610 |
| 3609 | 3610 | 3610 | 3611 | 3609 | 3611 | 3610 | 3608 | 3610 |
| 3607 | 3608 | 3606 | 3610 | 3607 | 3609 | 3609 | 3606 | 3610 |
| 3604 | 3605 | 3606 | 3605 | 3606 | 3605 | 3603 | 3603 | 3605 |
| 3612 | 3613 | 3611 | 3613 | 3613 | 3611 | 3613 | 3612 | 3612 |
| 3600 | 3600 | 3600 | 3594 | 3591 | 3586 | 3580 | 3577 | 3572 |
| 3608 | 3610 | 3609 | 3610 | 3609 | 3609 | 3608 | 3608 | 3608 |
| 3568 | 3564 | 3564 | 3561 | 3560 | 3558 | 3554 | 3551 | 3546 |
| 3608 | 3609 | 3607 | 3605 | 3607 | 3605 | 3607 | 3606 | 3607 |
| 3613 | 3612 | 3611 | 3613 | 3613 | 3612 | 3613 | 3612 | 3610 |
| 3611 | 3610 | 3608 | 3610 | 3609 | 3609 | 3610 | 3610 | 3610 |
| 3608 | 3609 | 3609 | 3611 | 3609 | 3609 | 3610 | 3609 | 3609 |
| 3580 | 3576 | 3571 | 3566 | 3563 | 3561 | 3556 | 3556 | 3554 |
| 3552 | 3549 | 3549 | 3550 | 3549 | 3546 | 3544 | 3543 | 3543 |
| 3612 | 3609 | 3609 | 3609 | 3607 | 3610 | 3609 | 3609 | 3609 |
| 3608 | 3610 | 3608 | 3610 | 3609 | 3607 | 3608 | 3608 | 3607 |
| 3611 | 3608 | 3611 | 3612 | 3607 | 3609 | 3607 | 3606 | 3609 |
| 3612 | 3613 | 3611 | 3611 | 3612 | 3612 | 3612 | 3613 | 3612 |
| 3613 | 3612 | 3612 | 3612 | 3612 | 3613 | 3612 | 3612 | 3613 |
| 3577 | 3568 | 3566 | 3565 | 3565 | 3564 | 3557 | 3554 | 3546 |

Figure 5: Data Sample from CsI(Tl) CSV File

## 3.2 Light Yield

The light yield was determined by summing the voltage counts for each signal and adjusting for the gain. Histograms of the total voltages were plotted, and a Gaussian curve was fitted to the peak corresponding to the full-energy absorption events. In order to account for different gains, we did as Page 5 shows to account for the differing values. These waveforms are then plotted to be fit in the followign form where $A$ is the amplitude, $\mu$ is the light yield, $\sigma$ is the deviation:

$$Ae^{\frac{(x-\mu)^2}{2\sigma^2}} \tag{5}$$

The following code schema demonstrates how the histogram is calculated:

```
# Step 1: Load the data from a CSV file
data = np.loadtxt("LYSO.csv", delimiter=",")

# Step 2: Apply gain correction
gain_correction = 20  # example gain value
data = data * (10 ** (-gain_correction / 20))

# Step 3: Calculate the baseline by averaging the first
    15 columns
baseline = np.average(data[:, :15], axis=1)

# Step 4: Compute the difference between the baseline
    and data
difference = np.expand_dims(baseline, axis=1) - data

# Step 5: Sum the differences to get the integrated
    waveform
integrated_waveform = np.sum(difference, axis=1)

# Step 6: Calculate the histogram
hist_values, bin_edges = np.histogram(
    integrated_waveform, bins=100)

# Step 7: Get the bin centers
bin_centers = (bin_edges[:-1] + bin_edges[1:]) / 2
```

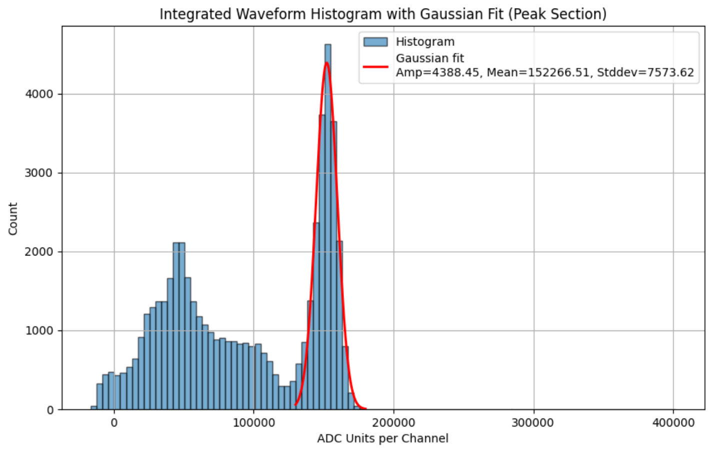


Figure 6: Light yield histogram for BGO crystal with Gaussian fit.

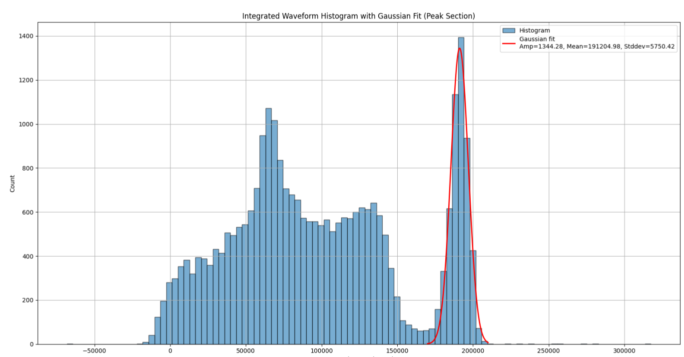


Figure 7: Light yield histogram for CsI(Tl) crystal with Gaussian fit.

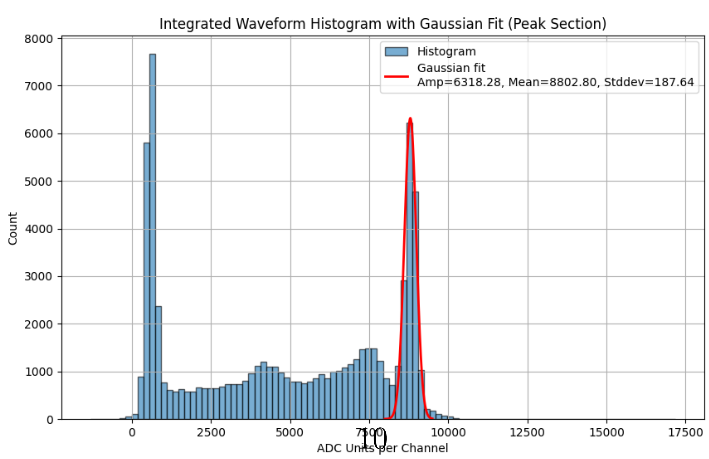




Figure 8: Light yield histogram for LYSO crystal with Gaussian fit.

In order to calculate the light yields of our substances, we must normalize the energy such that

$$\text{Light Yield (ADC units per keV)} = \frac{\mu}{662}$$

This will give us the light yield in ADC units per keV. Unfortunately due to the fact that we do not know the calibration factor of our system (which relates ADC units to photons). We will be using ratios to calculate our light yield.

The following are the light yield values in ADC units per keV from the experiment:

- **LYSO:** 13.30 ADC units per keV
- **BGO:** 5.78 ADC units per keV
- **CsI(Tl):** 23.00 ADC units per keV

## 3.3 Uncertainty of Light Yield

Table 3: Light Yield Ratios of CsI(Tl), LYSO, and BGO with Uncertainty

| **Ratio** | **Result with Uncertainty** |
|---|---|
| CsI(Tl)/LYSO | $1.73 \pm 0.15$ |
| LYSO/CsI(Tl) | $0.58 \pm 0.05$ |
| CsI(Tl)/BGO | $3.98 \pm 0.36$ |

The waveform data was measured in ADC channels. The voltages were summed, and each count in the histogram had an uncertainty of $\pm 0.5$ multiplied by the number of measurements in the signal, which remained consistent at 248 for all crystals. This value was used as a parameter for the `IMinuit` fitting process to calculate the error during curve fitting. For LYSO, the light yield uncertainty was $\pm 3.6$ ADC channels, and the fitted Gaussian curve resulted in a light yield of 13.30 ADC units per keV. For BGO, the time decay uncertainty was $\pm 4.0$ ADC channels, with a light yield of 5.78 ADC units per keV. The CsI(Tl) crystal had a light yield uncertainty of $\pm 5.2$ ADC channels, and the light yield was measured at 23.00 ADC units per keV.

## 3.4 Decay Time

Decay time refers to the time taken for the signal to decay to $1/e$ of its initial value. The average waveform for each crystal was calculated, and an exponential decay function was fitted to the falling edge:

$$V(t) = V_0 e^{-t/\tau}, \tag{6}$$

where $V(t)$ is the voltage at time $t$, $V_0$ is the initial voltage, and $\tau$ is the decay time constant.

The fitting range was selected from the peak voltage to the point where the signal approached the baseline. The range was determined using:

$$t_{\text{start}} = t_{\text{peak}}, \tag{7}$$

$$t_{\text{end}} = t_{\text{peak}} + \frac{1}{2}(t_{\text{final}} - t_{\text{peak}}). \tag{8}$$

The decay times for the crystals were measured as follows:

- **LYSO:** 67.75 ns
- **BGO:** 307.32 ns
- **CsI(Tl):** 935.14 ns

## 3.5 Uncertainty of Decay Time

Table 4: Measured Decay Times for CsI(Tl), BGO, and LYSO with Uncertainty

| **Crystal** | **Decay Time with Uncertainty (ns)** |
| --- | --- |
| CsI(Tl) | $935.14 \pm 52.00$ |
| BGO | $307.32 \pm 36.00$ |
| LYSO(Ce) | $67.75 \pm 7.30$ |

Each indivudal waveform had an uncertainty of $\pm 0.5$ ADC channels which is the raw uncertainty for each data point given the voltage being $55 \pm 0.5$.

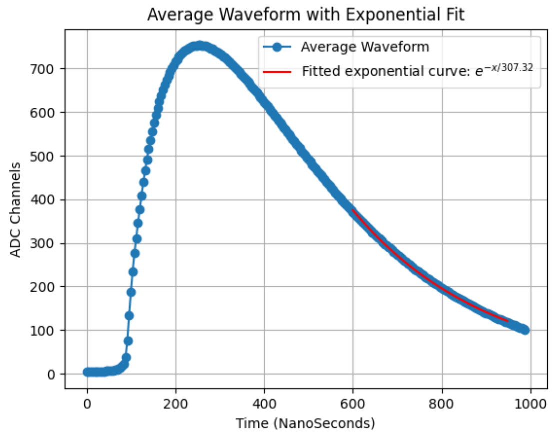


Figure 9: Decay time fit for BGO crystal.

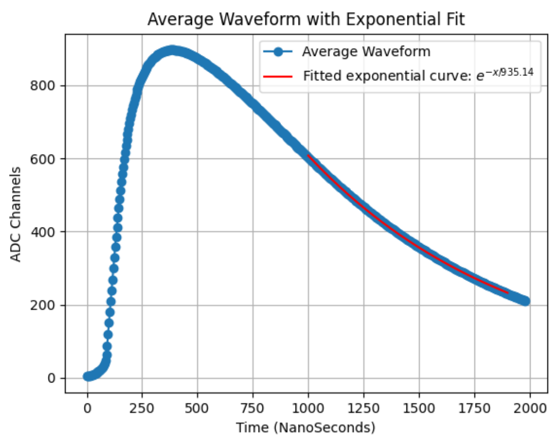


Figure 10: Decay time fit for CsI(Tl) crystal.

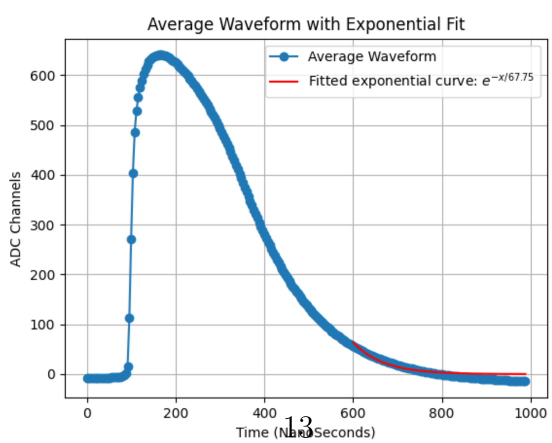




Figure 11: Decay time fit for LYSO crystal.

From there, we looked at the standard error of the mean (SEM) which is given by

$$\text{SEM} = \frac{\sigma}{\sqrt{N}}$$

$N = 248$ which are the number of waveforms thus the uncertainty will decrease by a factor of $\frac{1}{\sqrt{248}}$. We then used the IMinuit fitting package which will takes into account the uncertainty of each data point. The uncertainties in the decay time $\tau$ are derived from the fitting process.

# 4 Discussion

## 4.1 Comparison with Literature Values

The measured decay times and light yields are compared with accepted literature values in Tables 5 and 6, respectively.

Table 5: Comparison of Measured Decay Times with Literature Values

| **Crystal** | **Decay Time (ns)** | **Literature (ns)** | **Deviation (%)** |
|---|---|---|---|
| CsI(Tl) | $935.14 \pm 52.00$ | 1000 | 6.48% |
| BGO | $307.32 \pm 36.00$ | 300 | 2.44% |
| LYSO(Ce) | $67.75 \pm 7.30$ | 40 | 69.38% |

The deviations for CsI(Tl) and BGO are relatively small, falling within their uncertainty ranges and indicating good agreement with literature values. In the case of LYSO(Ce), the measured decay time deviates significantly from the accepted value by approximately 69.38% [6] [7] [8]. This large deviation likely reflects experimental limitations, as explained below.

## 4.2 Comparison of Light Yield Ratios with Literature Values

The calculated light yield ratios based on the experimental data are compared with the literature values in Table 6.

The ratio of CsI(Tl) to LYSO (1.73) is very close to the literature value of 1.7, showing excellent agreement within the uncertainty. However, the CsI(Tl)/BGO and LYSO/BGO ratios deviate significantly from the literature values by approximately 39.7% and 41.0%, respectively [9] [10]. These

Table 6: Comparison of Light Yield Ratios with Literature Values

| **Ratio** | **Measured Value** | **Literature Value** | **Deviation (%)** |
|---|---|---|---|
| CsI(Tl)/LYSO | $1.73 \pm 0.15$ | 1.7 | 1.76% |
| CsI(Tl)/BGO | $3.98 \pm 0.36$ | 6.6 | 39.7% |
| LYSO/BGO | $2.30 \pm 0.20$ | 3.9 | 41.0% |

deviations suggest systematic errors or experimental limitations in the BGO measurements.

## 4.3 Sources of Error

Several factors could have contributed to the discrepancies between the measured and literature values, especially for LYSO(Ce) and BGO:

**LYSO(Ce):**

The short decay time of LYSO(Ce) resulted in fewer data points before the signal decayed to the baseline, which made accurate fitting challenging. Overlapping signals and limited time resolution could have further introduced errors in the measurement. The large deviation (69.38%) in the decay time may also indicate that some additional effects, such as Compton scattering, may have contributed to the measured signal rather than just the photoelectric effect.

After the meeting discussing our results this Thursday, we discovered that the detector time resolution of the SiPM could have affected our result due to the finite time resolution of our SiPM. If the decay time of the scintillation light is shorter than the detector's time resolution, the results can underestimate or distort the decay time during measurements, thus leading to LYSO being measured to have a higher decay time than usual [11].

**BGO:**

The light yield measurements for BGO show significant deviations from accepted values. This could be due to lower signal-to-noise ratio and potential misalignment between the crystal and the SiPM. Environmental factors, such as temperature fluctuations, could have impacted the performance of the SiPM however our temperature data showcases that it could not have been the case.

From Thursday's meeting, we realized that there could have been potential issues from our data analysis based off of our graph. When calculating the integrated signal by summing `baseline - data` across each waveform, this approach does not account properly for the baseline offset from negative values or noise. Thus, this results in skewing the signal and affecting the histogram shape to give wrong light yields. Thus, the baseline should be determined for each waveform instead.

**General Sources of Error:**

Variations in the gain settings across different crystals could have affected the amplitude of the signals, leading to inconsistent results in the light yield measurements.

Due to the absence of an absolute calibration factor between ADC units and photons, the light yield was calculated using ratios rather than absolute values, which introduces an additional layer of uncertainty.

## 4.4 Uncertainty Analysis

The uncertainties in both decay times and light yields were propagated using the standard error of the mean (SEM), given by:

$$\text{SEM} = \frac{\sigma}{\sqrt{N}}$$

where $\sigma = 0.5$ ADC channels is the uncertainty in each measurement, and $N = 248$ is the number of waveforms averaged for each crystal. This uncertainty was further propagated through the fitting process using the `IMinuit` Python package, which provided the final uncertainties in the decay times and light yields.

The uncertainties are relatively small for CsI(Tl) and BGO in terms of decay times, and the light yield ratios for CsI(Tl) to LYSO show good agreement with literature values. However, the deviations in BGO's light yield and LYSO's decay time indicate that the measurement precision could be improved.

### 4.5 Improvements and Recommendations

In this experiment, we deepened our understanding of the three fundamental interactions in photon scattering: the photoelectric effect, Compton scattering, and pair production. We were surprised by the meticulous nature of the experimental process and the precision required in data collection and analysis. While the equipment was relatively easy to use after familiarization, troubleshooting consumed significant time due to various errors and technical issues. Despite these challenges, we would recommend this experiment due to its relevance in both particle and medical physics. Understanding the applications of CsI(Tl), BGO, and LYSO(Ce) crystals in various settings provides valuable insights into the practical uses of scintillation detectors.

LYSO crystals are used in medical imaging, particularly in Positron Emission Tomography or PET scans for short [12]. These are useful because of their high light yield and fast decay time as well as their good energy result. This allows for rapid detection of gamma ray results from positron annihilation events. BGO is also used in PET scans but also gamma cameras. Because of its slower decay time, it makes it efficient to detect high energy photons in high energy physics. For CsI(Tl), they can be used in many applications such as medical imaging, security screening, and astrophysics [1]. This makes CsI(Tl) one of the most versatile crystals.

I would definitely recommend this experiment to someone who's interested in medical physics but also just getting started in particle physics too.

## A Code Availability

All code used in this study is available on GitHub. Interested readers and researchers are encouraged to explore the repository to verify results, replicate experiments, or modify the code for further experimentation.

You can access the repository at the following link:

`https://github.com/MichaelCai2005/Crystal-Scintillation-`

# B Raw Data

This section provides an overview of the raw data files used in this experiment. All data files can be accessed through the following Google Drive link:

Google Drive Data Repository

# References


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